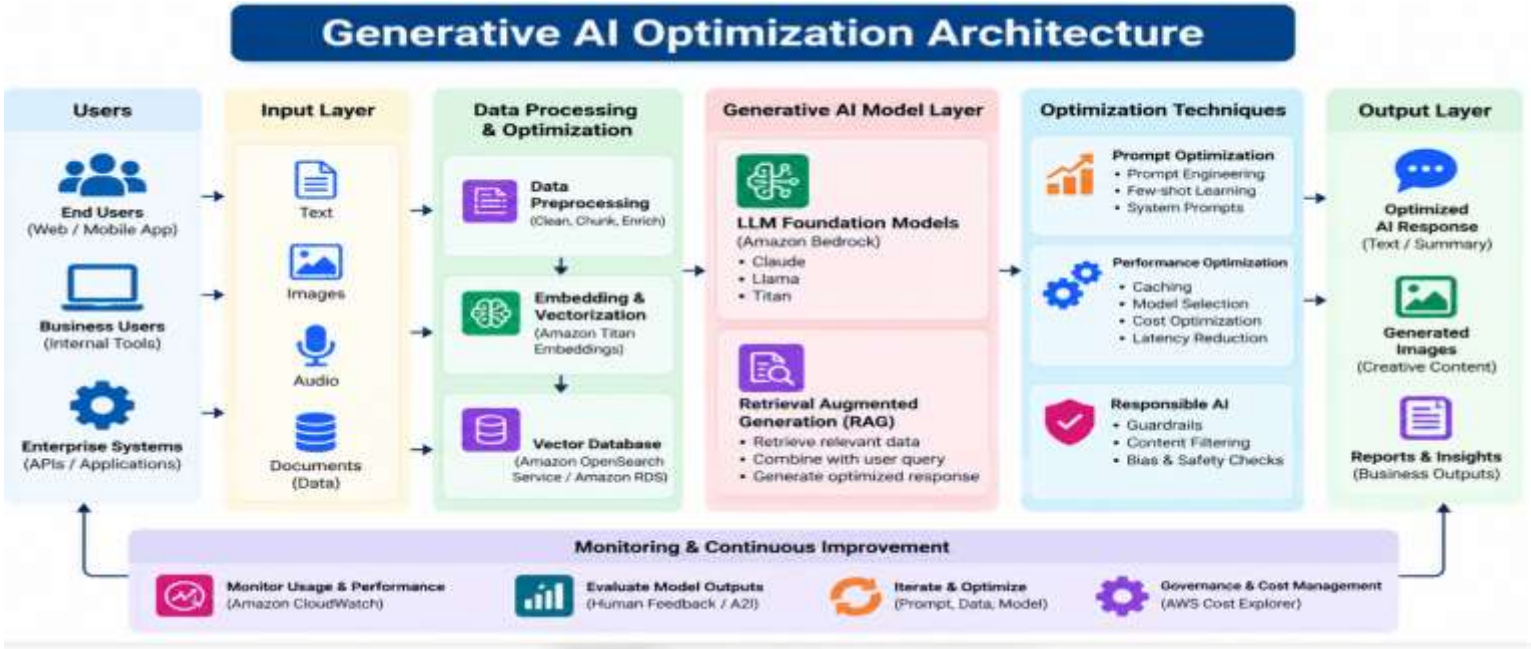
Generative AI Optimization – Token Efficiency, Cost & Resource Utilization

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- Faculty Advisor : Dr. Victor Govindaswamy

GENERATIVE AI OPTIMIZATION – INTRODUCTION

Main Course Topics Covered

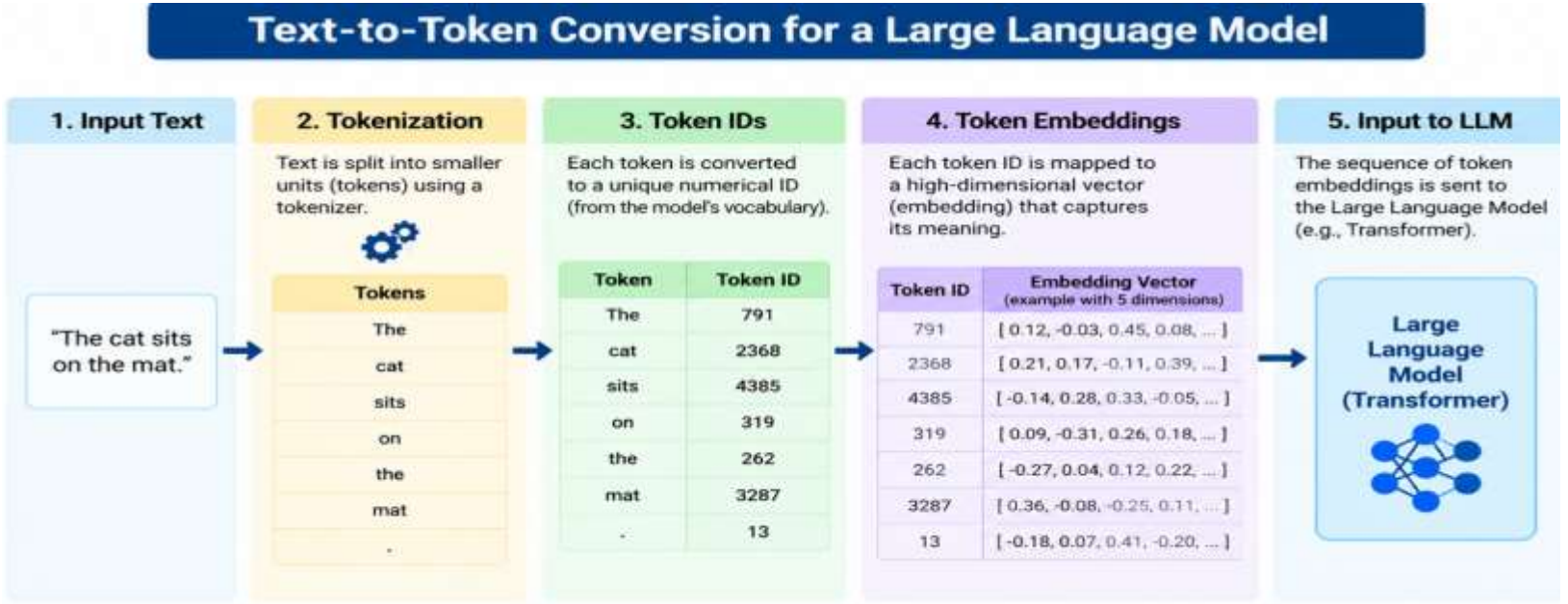
- Generative AI optimization
- Foundation Models
- Performance improvement
- Cost efficiency
- Resource utilization



Description

Generative AI optimization focuses on improving the performance, efficiency, and cost-effectiveness of AI applications. Developers can optimize model selection, token usage, and computing resources to build scalable and responsive Generative AI systems.

UNDERSTANDING TOKEN EFFICIENCY



Main Features

- Tokenization
- Input tokens
- Output tokens
- Token consumption
- Efficient prompt design

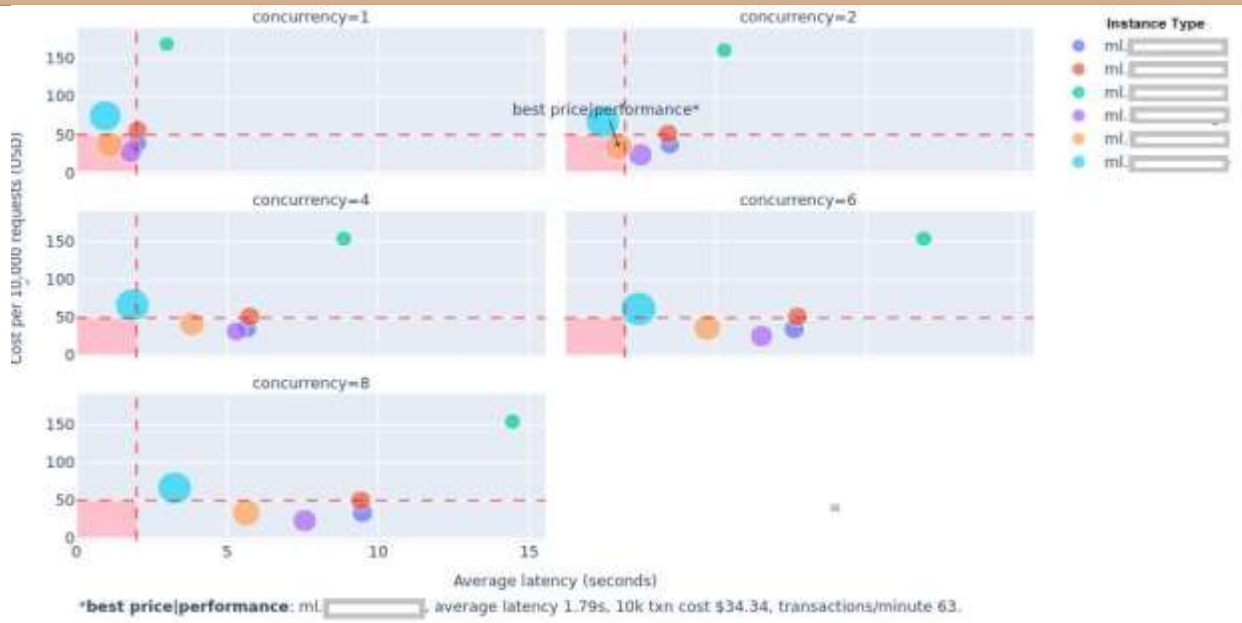
Description

Token efficiency involves managing the number of tokens processed by a Foundation Model. Understanding input and output tokens helps developers optimize prompts, manage context, and control inference costs while maintaining useful responses.

COST-EFFECTIVE MODEL SELECTION

Core Components

- Foundation Model selection
- Model capabilities
- Inference costs
- Response quality
- Application requirements



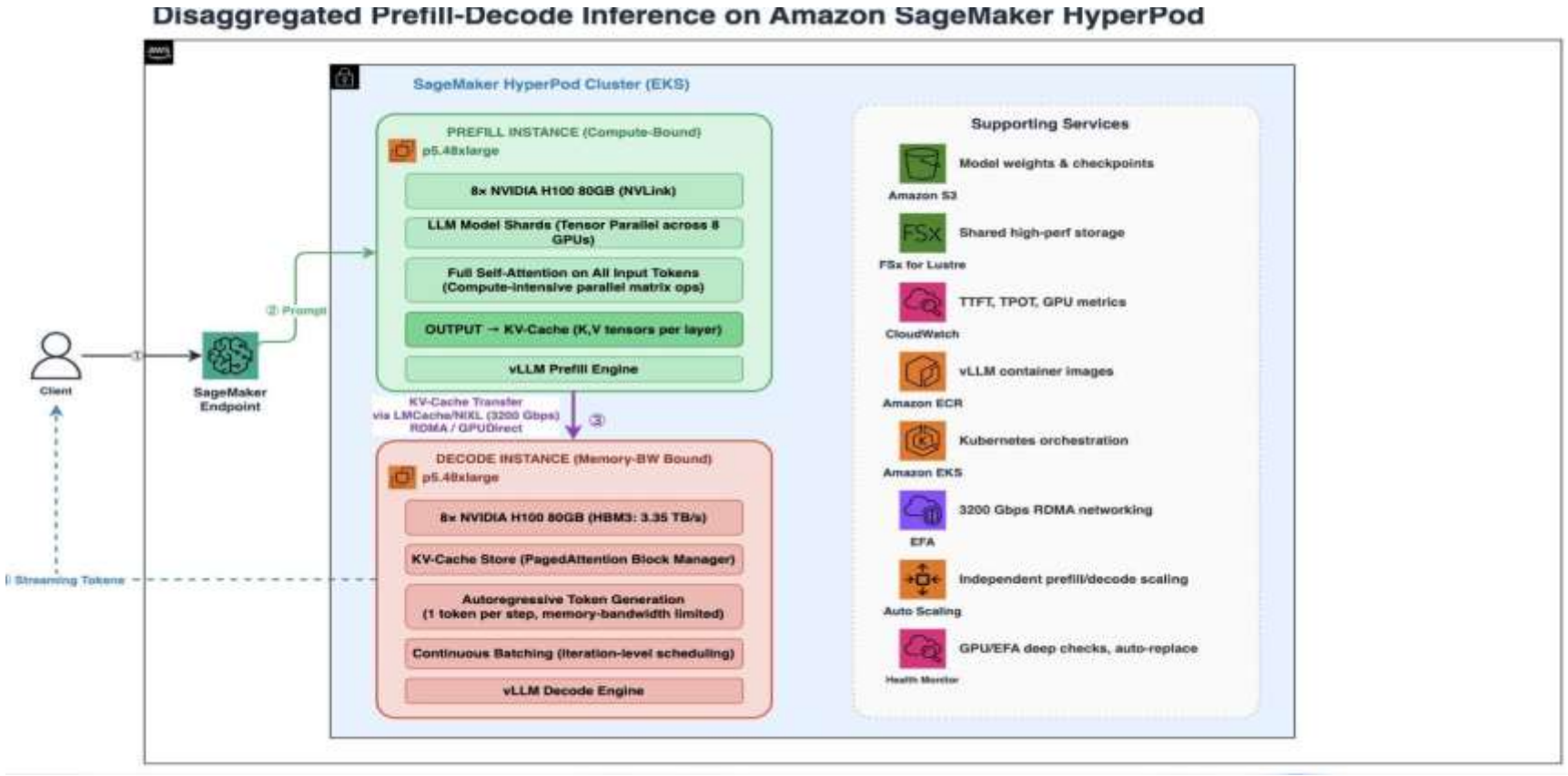
Description

Selecting a cost-effective Foundation Model requires evaluating its capabilities, performance, pricing, and suitability for a particular task. Choosing a model that meets application requirements can help balance response quality and operational costs.

MANAGED SERVICE FOR APACHE FLINK – HANDS-ON

Important Features

- Hands-on experience
- Streaming application
- Real-time processing
- Flink configuration
- Data analysis



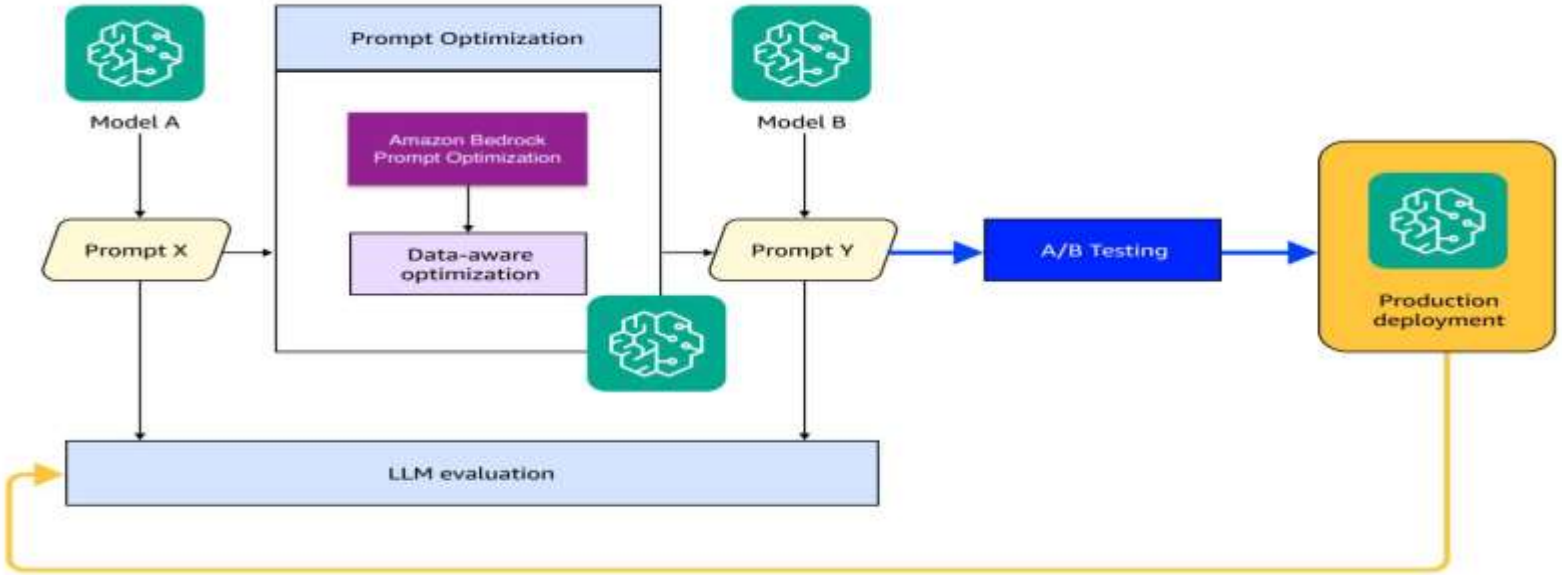
Description

The hands-on lesson provides practical experience with Amazon Managed Service for Apache Flink. It demonstrates how streaming data can be processed and analyzed using a managed AWS environment.

MAXIMIZING RESOURCE UTILIZATION AND THROUGHPUT

Main Points

- Resource utilization
- Request processing
- Throughput optimization
- Concurrent requests
- Workload management



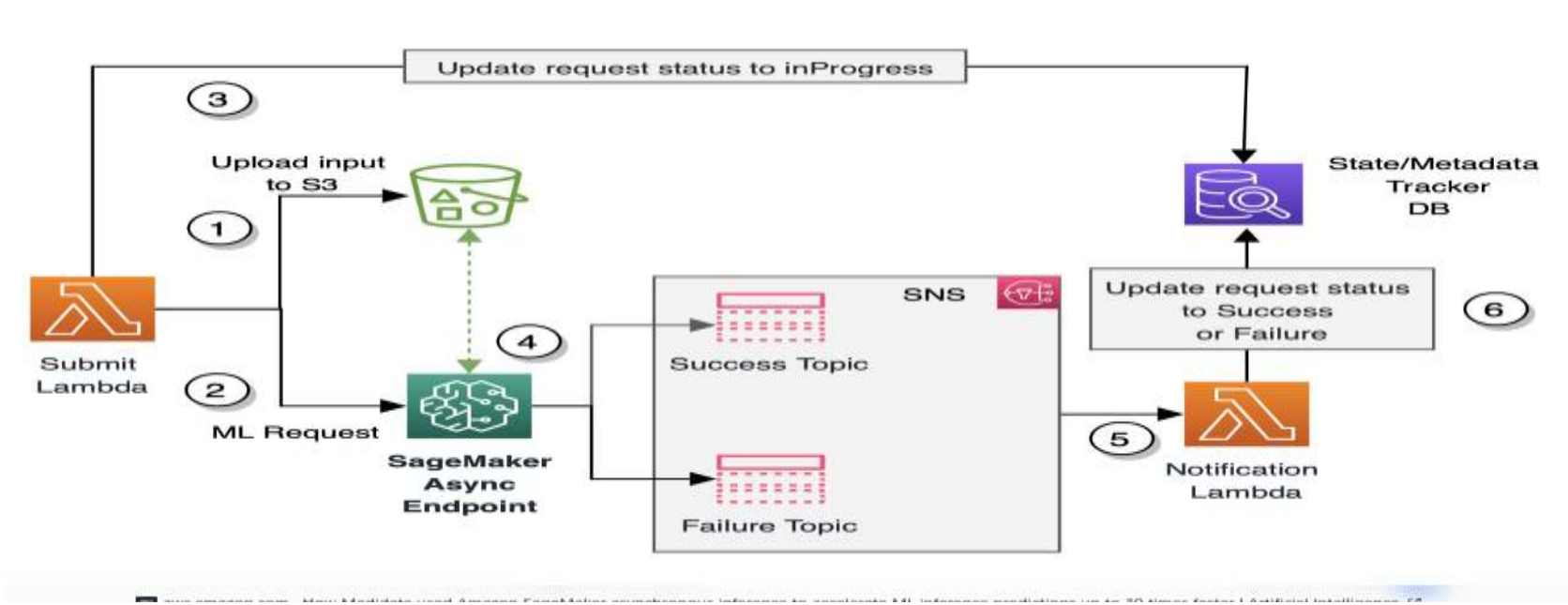
Description

Resource utilization and throughput are important factors in Generative AI application performance. Efficient workload management and request processing can help applications handle more requests while making effective use of available computing resources.

PROMPT AND CONTEXT OPTIMIZATION

Main Points

- Prompt engineering
- Context management
- Relevant information
- Token reduction
- Response quality



Description

Prompt and context optimization involves providing clear instructions and relevant information to a Foundation Model. Removing unnecessary content and organizing prompts effectively can reduce token consumption and help the model focus on the task.

EFFICIENT AI INFERENCE WORKLOADS

Main Features

- AI inference
- Request management
- Concurrent processing
- Processing efficiency
- Workload optimization



Description

AI inference workloads involve processing requests through a deployed Foundation Model. Efficient request management and appropriate workload configurations can help improve processing efficiency and application responsiveness.